I.16 · Put an interior angle inside an exterior angle

If one side of any triangle is extended, the exterior angle is greater than either of the interior and opposite angles.

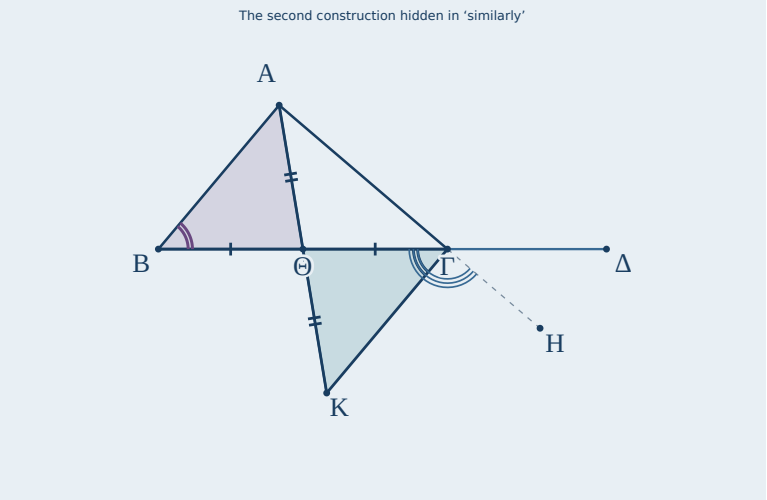


Plate at step 9: Bisect BΓ and repeat for the other interior angle.

The second construction hidden in ‘similarly’

Now bisect BΓ at Θ, extend AΘ to K with ΘK = ΘA, and join KΓ. Θ and K are modern labels expanding the source's abbreviated instruction. The copied angle lies inside BΓH, which is vertically opposite and equal to AΓΔ.

The repeated side–angle–side argument

ΒΘΑ	ΓΘΚ	Correspondence
ΒΘ	ΓΘ	equal halves · I.10
∠ΒΘΑ	∠ΓΘΚ	vertical angles · I.15
ΘΑ	ΘΚ	equal extension
∠ΑΒΘ	∠ΚΓΘ	corresponding angles · I.4

∠ΑΓΔ > ∠ΒΑΓ (whole > proper part) · ΒΘ = ΘΓ; ΑΘ = ΘΚ; opposite angles at Θ are equal · ∠ΑΒΓ = ∠ΘΓΚ < ∠ΒΓΗ = ∠ΑΓΔ

The proposition

Παντὸς τριγώνου μιᾶς τῶν πλευρῶν προσεκβληθείσης ἡ ἐκτὸς γωνία	In quouis triangulo uno latere producto angulus extrinsecus	If one side of any triangle is extended, the exterior angle is

ἑκατέρας τῶν ἐντὸς καὶ ἀπεναντίον γωνιῶν μείζων ἐστίν.	positus utrouis angulo interiore et opposito maior est.	greater than either of the interior and opposite angles.

1. Extend ΒΓ past Γ to Δ Given

Ἔστω τρίγωνον τὸ ΑΒΓ, καὶ προσεκβεβλήσθω αὐτοῦ μία πλευρὰ ἡ ΒΓ ἐπὶ τὸ Δ· λέγω, ὅτι ἡ ἐκτὸς γωνία ἡ ὑπὸ ΑΓΔ μείζων ἐστὶν ἑκατέρας τῶν ἐντὸς καὶ ἀπεναντίον τῶν ὑπὸ ΓΒΑ, ΒΑΓ γωνιῶν.	Sit triangulus ΑΒΓ, et producatur unum latus eius ΒΓ ad Δ punctum. dico esse ∠ ΑΓΔ > ΓΒΑ et ΑΓΔ > ΒΑΓ.	Let ΑΒΓ be a triangle and extend its side ΒΓ to Δ. I claim that exterior angle ΑΓΔ is greater than each opposite interior angle ΓΒΑ and ΒΑΓ.

ἑκατέρας / utrouis means each of the two separately. The claim is an inequality with each interior angle, not an equality with their sum. The adjacent interior angle at Γ is excluded by ἀπεναντίον / opposito.

Reason: Postulate 2 · extend a segment A finite straight line may be continued in a straight line. In I.2 the extensions go away from Δ, past A and B. No new length is measured here.

Printed source: Greek pp. 42; Latin pp. 43.

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2. Bisect ΑΓ at E Construct

Τετμήσθω ἡ ΑΓ δίχα κατὰ τὸ Ε,	secetur ΑΓ in duas partes aequales in E,	Bisect ΑΓ at E.

τετμήσθω / secetur commands a cut; δίχα / in duas partes aequales specifies equal halves. This is I.10, now available as a proved construction.

Reason: I.10 · bisect the side ΑΓ Construct the midpoint E with I.10. The plate computes it by intersecting the angle bisector with the given segment, rather than assigning an arithmetic midpoint as a construction permission.

Printed source: Greek pp. 42; Latin pp. 43.

3. Join, extend, copy, and join again Construct

καὶ ἐπιζευχθεῖσα ἡ ΒΕ ἐκβεβλήσθω ἐπ᾽ εὐθείας ἐπὶ τὸ Ζ, καὶ κείσθω τῇ ΒΕ ἴση ἡ ΕΖ, καὶ ἐπεζεύχθω ἡ ΖΓ, καὶ διήχθω ἡ ΑΓ ἐπὶ τὸ Η.

et ducta BE producatur in directum ad Z, et ponatur EZ = BE, et ducatur ΖΓ, et educatur ΑΓ ad H.

Join BE and extend it straight to Z, laying off EZ equal to BE. Join ΖΓ and continue ΑΓ to H.

ἐπιζευχθεῖσα / ducta gives the joining as already done before the extension command. ἐκβεβλήσθω, κείσθω, ἐπεζεύχθω and διήχθω form a sequence of distinct operations, not one instruction to draw the final shape.

Reason: Extend BE and lay off EZ equal to BE Continue BE beyond E and choose Z with EZ = BE; join ΖΓ and extend ΑΓ past Γ to H. The source gives these commands without a separate citation for copying the length. Here I.2 supplies a length at E, then a circle at E cuts the continuation. The copied length is not silently assumed to be a new compass permission.

Printed source: Greek pp. 42; Latin pp. 43.

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4. Collect the two equal side pairs Prove

Ἐπεὶ οὖν ἴση ἐστὶν ἡ μὲν ΑΕ τῇ ΕΓ, ἡ δὲ ΒΕ τῇ ΕΖ, δύο δὴ αἱ ΑΕ, ΕΒ δυσὶ ταῖς ΓΕ, ΕΖ ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ·

iam quoniam AE = EΓ et BE = EZ, duae rectae AE, EB duabus ΓE, EZ aequales sunt altera alteri.

AE equals EΓ, and BE equals EZ. Thus the two sides AE, EB equal ΓE, EZ respectively.

ἑκατέρα ἑκατέρᾳ / altera alteri keeps the correspondence ordered: AE with ΓE, EB with EZ. It does not say all four lengths are equal.

Reason: The midpoint and the copied length AE = EΓ from I.10, and BE = EZ by construction. These are the two corresponding side pairs of ABE and ΓZE. The angle needed next is the one included between them.

Printed source: Greek pp. 42; Latin pp. 43.

5. Use I.15 at the crossing E Prove

καὶ γωνία ἡ ὑπὸ ΑΕΒ γωνίᾳ τῇ ὑπὸ ΖΕΓ ἴση ἐστίν· κατὰ κορυφὴν γάρ·	et ∠ AEB = ZEΓ (nam ad uerticem eius est).	Angle AEB equals ZEΓ, for they are vertically opposite.

γάρ / nam supplies the reason, compressed into ‘vertically opposite’. The previous lesson turns that short phrase into a proved equality. E is the vertex of both included angles.

Reason: I.15 · the included angles are vertically opposite The lines AΓ and BZ cross at E. Their vertically opposite angles AEB and ZEΓ are equal by I.15. We now have two sides and their included angle, not two sides and an arbitrary angle.

Printed source: Greek pp. 42; Latin pp. 43.

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6. Apply side–angle–side Prove

βάσις ἄρα ἡ ΑΒ βάσει τῇ ΖΓ ἴση ἐστίν, καὶ τὸ ΑΒΕ τρίγωνον τῷ ΖΕΓ τριγώνῳ ἐστὶν ἴσον, καὶ αἱ λοιπαὶ γωνίαι ταῖς λοιπαῖς γωνίαις ἴσαι εἰσὶν ἑκατέρα ἑκατέρᾳ, ὑφ᾽ ἃς αἱ ἴσαι πλευραὶ ὑποτείνουσιν·	itaque basis AB basi ZΓ aequalis est et △ ABE = ZEΓ, et reliqui anguli reliquis aequales sunt alter alteri, sub quibus aequalia latera subtendunt.	Therefore AB equals ZΓ, the triangles ABE and ZEΓ are equal, and their remaining angles are equal respectively where the equal sides subtend them.

λοιπαί / reliqui now means the remaining angles after the included one. The source's triangle naming does not list vertices in matching correspondence; use the stated side pairs to read the correspondence A ↔ Γ, B ↔ Z, E ↔ E.

Reason: I.4 · transfer the opposite interior angle The two side pairs and the equal included angles give AB = ZΓ and equality of the triangles and remaining corresponding angles by I.4. In particular BAE = EΓZ. This places a copy of the interior angle inside the exterior angle.

Printed source: Greek pp. 42; Latin pp. 43.

7. Identify the copied interior angle Prove

ἴση ἄρα ἐστὶν ἡ ὑπὸ ΒΑΕ τῇ ὑπὸ ΕΓΖ.	itaque ∠ BAE = EΓZ.	In particular, BAE equals EΓZ.

ἄρα / itaque draws out the one corresponding angle needed next. Because E lies on AΓ, BAE is the original interior angle BAΓ.

Reason: I.4 · transfer the opposite interior angle The two side pairs and the equal included angles give AB = ZΓ and equality of the triangles and remaining corresponding angles by I.4. In particular BAE = EΓZ. This places a copy of the interior angle inside the exterior angle.

Printed source: Greek pp. 42; Latin pp. 43.

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8. The exterior angle contains that angle as a proper part Prove

μείζων δέ ἐστιν ἡ ὑπὸ ΕΓΔ τῆς ὑπὸ ΕΓΖ· μείζων ἄρα ἡ ὑπὸ ΑΓΔ τῆς ὑπὸ ΒΑΕ.	uerum ∠ EΓΔ > EΓZ. quare ∠ AΓΔ > BAE.	But EΓΔ is greater than its part EΓZ. Therefore exterior angle AΓΔ is greater than BAE.

μείζων with the genitive τῆς means ‘greater than’. The sentence crosses the page in both languages. EΓΔ and AΓΔ name the same exterior angle because E lies on ΓA.

Reason: Common notion 5 · the exterior contains a proper part Ray ΓZ lies strictly inside exterior angle EΓΔ in this nondegenerate plane triangle. Thus EΓΔ is larger than EΓZ, which equals BAE. The strict containment, not an assumed triangle-angle sum, supplies the inequality.

Printed source: Greek pp. 42, 44; Latin pp. 43, 45.

9. Bisect ΒΓ and repeat for the other interior angle Prove

Ὁμοίως δὴ τῆς ΒΓ τετμημένης δίχα δειχθήσεται καὶ ἡ ὑπὸ ΒΓΗ, τουτέστιν ἡ ὑπὸ ΑΓΔ, μείζων καὶ τῆς ὑπὸ ΑΒΓ.	similiter recta ΒΓ in duas partes aequales secta demonstrabitur etiam ∠ ΒΓΗ > ΑΒΓ, h. e. ∠ ΑΓΔ > ΑΒΓ.	Similarly, after bisecting ΒΓ, ΒΓΗ will be shown greater than ΑΒΓ too. That is, ΑΓΔ is greater than ΑΒΓ.

ὁμοίως / similiter calls for a second construction, not merely a second numerical check of the first one. τουτέστιν / h. e. identifies ΒΓΗ with the equal vertically opposite angle ΑΓΔ. The plate's Θ and K make the abbreviated second construction visible; those labels are our additions.

Reason: Repeat the construction for the other interior angle The source says to bisect ΒΓ and reason similarly. The plate makes that omitted drawing explicit: Θ is the midpoint of ΒΓ, and ΑΘ is extended to K with ΘK = ΘA. These two labels are modern additions. I.15 and I.4 give ΘΓK = ΑΒΓ inside ΒΓΗ; I.15 also gives ΒΓΗ = ΑΓΔ. The same exterior angle is therefore larger than the other opposite interior angle too.

Printed source: Greek pp. 44; Latin pp. 45.

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10. The exterior is greater than each opposite interior angle Conclude

Παντὸς ἄρα τριγώνου μιᾶς τῶν πλευρῶν προσεκβληθείσης ἡ ἐκτὸς γωνία ἑκατέρας τῶν ἐντὸς καὶ ἀπεναντίον γωνιῶν μείζων ἐστίν· ὅπερ ἔδει δεῖξαι.	Ergo in quouis triangulo uno latere producto angulus extrinsecus positus utrouis angulo interiore et opposito maior est; quod erat demonstrandum.	Therefore, when one side of any triangle is extended, the exterior angle is greater than each of the opposite interior angles. This is what had to be shown.

παντός / quouis makes the claim general. The proof uses the earlier constructions, I.4, I.15 and the whole–part principle. It has not used the parallel postulate or a triangle-angle-sum theorem.

Reason: The construction is complete The requested object has been constructed and its defining equality has been proved. The closing phrase says that the task was to make something, rather than merely state a theorem.

Printed source: Greek pp. 44; Latin pp. 45.

A strict inequality before the parallel postulate

I.16 proves that the exterior angle is greater than either opposite interior angle. It does not yet prove the later equality with their sum. The key move is construction: copy one interior angle into a proper part of the exterior, then repeat with the other side bisected. At step 9 the plate draws that second construction explicitly.

Definitions and starting permissions used in this proof

Postulate 2 · β′ / II · Extend a straight line

Καὶ πεπερασμένην εὐθεῖαν κατὰ τὸ συνεχὲς ἐπ’ εὐθείας ἐκβαλεῖν.	Et ut recta linea terminata in directum educatur in continuum.	And that a finite straight line can be extended continuously in a straight line.

πεπερασμένην / terminata identifies a bounded segment. ἐπ’ εὐθείας / in directum keeps the extension straight: the command does not permit a turn at the endpoint.

Printed source: Greek pp. 8; Latin pp. 9.

Common notion 5 in the usual list · η′ / VIII in Heiberg · Whole and proper part

Καὶ τὸ ὅλον τοῦ μέρους μεῖζόν [ἐστιν].	Et totum parte maius est.	And the whole [is] greater than the part.

Greek uses the genitive τοῦ μέρους after the comparative μεῖζον; Latin uses the ablative parte after maius. Both mean ‘greater than the part’. The brackets around [ἐστιν] belong to the printed Greek.

Printed source: Greek pp. 10; Latin pp. 11.

Greek and Latin: Euclidis opera omnia, vol. I, ed. and Latin translation J. L. Heiberg, Leipzig: Teubner, 1883, printed pp. 2–45. Public domain. University of Toronto scan. English, reading notes and redrawn coordinate diagrams: Stoicheia (2026), MIT. Greek ϑ is normalized to θ; line breaks and inline references are omitted. Latin u/v spelling is retained. Heiberg’s square brackets in I.1, I.4, I.8, I.13 and I.15 are retained. Printed angle and triangle symbols are represented as ∠ and △. The English is a reading translation, including explicit explanations of implied intersections.

The diagram is numerical evidence, not a formal proof. Length units and orientation are modern teaching choices.